Sandeep K. Juneja

Professor of Computer Science,
Director, Safexpress Centre for Data, Learning and Decision Sciences, Ashoka University
sandeep.juneja2010@gmail.com, 91-9967932124

Academic Experience

03/24 - present	Ashoka University. Computer Science Department. Professor and Director of Centre for Data, Learning and Decision Sciences. Centre is pursuing an active agenda on weather and climate and financial modelling as well as research on Generative AI methods and sequential learning theory.
02/26 - present	Indian Institute of Technology Bombay. Department of Electrical Engineering. Visiting Professor
12/02 – 8/24	Tata Institute of Fundamental Research, School of Technology and Computer Science, Mumbai. Senior Professor (Jan 2021- Aug 24), Dean (Jan 2017 – Sep 2021, Sept 2023 – January 2024), Professor (Jul 2011 – Dec 2020), Associate Professor (2004-11), Reader (2002-04).
11/96 – 12/03	Indian Institute of Technology Delhi, Industrial Eng. Group, Dept. of Mech. Eng., Associate Professor (2002); Assistant Professor (1997 – 2002)

Corporate experience before joining academia (after PhD)

9/95 - 10/96	Andersen Consulting, New Delhi. Senior Consultant
9/93 - 6/95	American Credit Indemnity (Company of Dun & Bradstreet Corporation) Baltimore, MD. Specialty credit insurance company. Director Quantitative Analysis

Education

89-93	Ph.D., Operations Research, Stanford University
89-92	M.S., Statistics, Stanford University
85-89	B.Tech., Mech. Eng., Indian Institute of Technology Delhi

Selected visiting and adjunct academic positions

July 22 – July 23	Visiting Researcher, Google Research, Bangalore India
March 23 (1 wk)	Climate Modelling Alliance, Caltech
Jan 22 – June 22	Visiting Professor, Computer Science, Ashoka University
Mar 16 – Feb 17	Adjunct academic, CAFRAL (Centre for Advanced Financial Research and Learning). Research wing of Reserve Bank of India. Mumbai
Sep-Dec 15	Senior Academic Fellow, CAFRAL, Research wing of Reserve Bank of India. Mumbai
Nov 13 (1 wk)	Department of Applied Mathematics, Ecole Polytechnique, Palaiseau,
May 15 (1 wk)	Cedex, France
Oct-Nov 12	ICERM (Institute for Computational and Experimental Research in Mathematics) Brown University

Nov 11 (2 wk)	Korteweg-de Vries Institute for Mathematics, University of Amsterdam, Visiting Professor on the STAR Grant
Oct 10, Apr-May 11	School of Mathematical and Computer Sciences, Heriot Watt University, Visiting Fellow on EPSRC Grant
June 10 (1wk)	Newton Institute of Mathematical Sciences, Cambridge University, UK. Invited guest in a program on Stochastic Network.
Summers 04, 05, 07	Graduate School of Business, Columbia University
Summers 97, 98, 99, 04, 05, 07	Industrial Engineering and Operations Research, Columbia University
3/06 – 12/06	Indian School of Business, Hyderabad
1/02 – 4/02	Tata Institute of Fundamental Research, School of Technology and Computer Science, Mumbai
5/01 – 6/01	University of Twente, Netherlands. Computer Science Department
1/2000 – 4/2000	Stanford University, Department of Management Science and Engineering
7/03 – 12/05	Indira Gandhi Institute for Developmental Research, Mumbai. Adjunct Associate Professor

Publications The list is attached separately

Editorial boards, professional service

2024 – Present	Area Editor, Operations Research (in simulation area)
2017 – present	Associate Editor, Stochastic Systems
2008 – 2016	Associate Editor, Mathematics of Operations Research
2003-2009	Associate Editor, Management Science. In the area of Stochastic Models and Simulation
2007 - 2010	Associate Editor, ACM TOMACS. In the area of Monte Carlo simulation
2026	Member, Advisory Board. IGIDR for formation of Department of Statistics and Data Science
2024 – present	Member, Technical Advisory Group, Collaboratory, a laboratory for pandemic and epidemic intelligence within the World Health Organization (WHO) Hub.
2021 – 2024	Chair (Dec 21- Dec 22), Member (Dec 20 - Dec 23), INFORMS Applied Probability Society Prize Selection Committee
2021	Member of Committee for modelling of medical oxygen requirements across states and UT. Under Niti Aayog
2021 – 2022	Member, Lancet Covid-19 Commission India Task Force
2022 – present	Member, Board of Studies, Artificial Intelligence and Machine Learning department, Dwarkadas J. Sanghvi College of Engineering, Mumbai
2019 – 2022	Member, Bombay Chamber of Commerce, Banking, Finance and Economics Committee
2018 – 2022	Member, Board of Studies, School of Mathematical Sciences at the NMIMS, Mumbai
2016 – 2018	Member, National Advisory Board, Economic Sciences at IIT Kanpur

2016 – 2019	Member, Academic Council, Indira Gandhi Institute for Developmental Research, Mumbai
2008 - 2010	Guest Editor for a special issue of Annals of Operations Research
2006, 2008	Track Coordinator, Risk Analysis, Winter Simulation Conference

Program Committee Member

- Senior PC member COLT 2024, ACM Sigmetrics (2021- 22), INFORMS Nov. 2018, INFORMS Applied Probability Conference, July 2017, Scientific Committee Member, Statistical Methods in Finance Conference, CMI 2016, INFORMS Applied Probability Conference 2015, IFIP Performance 2015

Awards and recognition

- Finalist for Best Theoretical Paper at Winter Simulation Conference 2022 for the paper Exact Optimal Fixed-width Confidence Interval Estimation for the Mean.
- Wockhardt Foundation Shining Star Award. January 2022
- Best paper award at CRISIL Doctoral Symposium 2017 won by graduate student Anand Deo for joint work on Calibration of Credit Default Probabilities in Discrete Default Intensity and Logit Models
- Best paper award at the ICST Sixth International Conference on Performance Evaluation Methodologies and Tools (Valuetools 2012) for the paper The Concert Queueing Game with Random Arrivals Volume
- Ranked amongst the most productive researchers in management from India (for period 1990-2009) in a study conducted by researchers from Aditya Birla Centre at the London Business School
  http://www.peerpower.com/et/3464/Taking-stock-of-Indian-management-research
- Best paper award at the ICST Fourth International Conference on Performance Evaluation Methodologies and Tools (Valuetools 2009) for the paper The Concert/Cafeteria Queuing Problem: A Game of Arrivals
- Recipient of Yahoo Academic Research Grant for the year 2009-10
- Faculty Partnership award from IBM Research Lab Yorktown Heights for the year 2001-2002
- First Patent Invention Award, IBM Research Lab India, 2000
- Fellowship and continued support, Department of Operations Research, Stanford University, 1989-93
- Merit scholarship for topping in the Department of Mechanical Engineering in fourth and fifth semesters (1987-88), IIT Delhi

Former PhD students

1. Santanu Dey. 2013. Goldman Sachs, 2013 – 15. Unfortunately, Santanu passed away in 2015
   Optimal Change of Measure for Model Selection and Efficient Simulation of Rare Event Probabilities with Financial Applications
2. Ankush Agarwal. 2015. Faculty at Adam Smith Business School, Glasgow University
   Monte Carlo Based Methods for Pricing American Options
3. Karthyek RA Murthy. 2015. Faculty at Singapore University of Technology and Design.
   Rare Events in Heavy Tailed Stochastic Systems: Algorithms and Analysis
4. Tushar Raheja. 2016. (jointly with Prof. Kiran Seth, IITD). Popular writer, film maker

Modeling traffic congestion – to wait or to be late

5. Sarat Babu Moka. 2017. Faculty at University of New South Wales, Sydney.
   Invariant Measures for Queueing and Spatial Markov Processes: Algorithms and Analysis
6. Anand Deo. 2021. Faculty at IIM Bangalore.
   An Asymptotic Study of Risk in Financial Systems - Algorithms and Analysis
7. Shubhada Agrawal. 2022. Post-doc at Georgia Tech.
   Bandits with Heavy Tails: Algorithms, Analysis & Optimality
8. Anirban Bhattacharjee. 2026. Rare Events in Stochastic Multi-Armed Bandits
9. Sushant Vijayan. 2026. Sequential Decision-Making under Uncertainty and Constraints: A Synthesis of Differential Games and Stochastic Bandits

Current PhD students

1. Agniv Bandopadhyay (TIFR)
2. Sai Rakshith S
3. Ojas Kumar
4. Keshav Kumar

Post-doctoral visitors mentored

- Kushal Bhowmick. July 26 – present. At SCDLDS Ashoka University (PhD HRI)
- Sarvesh Iyer. 2024 - present. At SCDLDS Ashoka University (PhD ISI Bangalore)
- Sagar Shrivastava. Aug-Dec 2025. At SCDLDS Ashoka University (PhD TIFR)
- Poonam Kesarwani. Jan. 2020 – Nov. 2021. (PhD IIT Kanpur)
- Subhashini Krishnasamy. September 2017 - November 2018. (PhD UT Austin)
- Vineeth Chintala April - June 2017. (PhD TIFR Mathematics, Mumbai)
- Tejas Bordas. June - September 2016. (PhD IIT Bombay)

Doctoral committees

- Opponent in PhD thesis defence of Newton Mwai Kinyanjui. October 2025. Dept. of CS, Chalmers University. Sweden
- Member thesis review committee, Pai-An Wang, KTH Stockholm. September 2024.
- Member thesis review committee, Marc Jourdain, University of Lille, France. June 2024.
- Member doctoral committee for Fatale Santosh S, EE, IIT Bombay. Jan 2024
- Member doctoral committee for Vikas Deep, 2023. Northwestern University.
- External Examiner for PhD Thesis of Prasenjit Karmaker, CS Dept., IISc. 2018
- External Examiner for PhD Thesis of Ajin George Joseph, CS Dept., IISc. 2017
- External Examiner for Phd Thesis of Tejas Bodas, EE Dept., IIT Bombay. April 2016
- Opponent in PhD thesis defence of Pierre Nyquist. Department of Mathematics, KTH Stockholm. 2014
- Chaired the PhD thesis defence committee for V Sasidevan, TIFR. 2014
- External Examiner for PhD thesis of Chandan Pal. Dept. Mathematics, IIT Bombay. 2013
- External Examiner for PhD thesis of Tamal Banerjee. 2013. Dept. Mathematics, IISc Bangalore
- Examiner for PhD thesis Samarth Chandra. 2009. Dept. of Theoretical Physics, TIFR

- External Examiner for PhD thesis of Ozgur Kaya. 2005. Columbia Business School
- External Examiner for PhD thesis of Z. Huang. 2004. Department of Industrial Engineering and Operations Research, Columbia University
- Examiner for PhD thesis of Rahul Jain. 2004. School of Technology and Computer Science, TIFR

Key consultancies and industry interaction

July 22 – July 23	Visiting Researcher, Google Research, Bangalore India
Sep. - Nov. 20	Taught a course on stochastic calculus and math finance to quant associates in Bank of America (BA Continuum), Mumbai
Aug. 19	Taught a course on math finance to quant associates in JP Morgan, Mumbai
Mar. 19	Taught a short course on equity and interest rate financial derivatives to quants at Ernst and Young in Bangalore
Sep – Dec 14 Feb – Apr 16	Taught introductory math finance to quants at Nomura Bank, Mumbai
Aug. 14	Taught Interest Rate Models to quants at Credit-Suisse Bank, Mumbai
Sep 11 – July 12	Capital Metrics and Risk Solutions, Pune, India. Helping in designing a cutting edge portfolio risk measurement product
Aug - Dec 10	Taught introductory math finance to quants at ICICI Bank, Mumbai
July 10	Taught introductory math finance to quants at Morgan Stanley in Mumbai
Apr – Sep 10	Taught cutting edge interest rate models, stochastic volatility based models, credit risk models to the advanced quants at Nomura, Mumbai
June 09	Yahoo Research Lab, Bangalore. Designed algorithms for better estimation of click probabilities of Internet advertisements
Jan – Sep 08	Bank of America. (On leave from TIFR) Vice President, Head Quantitative Analysis for Bank of America Continuum Solutions at Mumbai. Amongst the fifteen members of the executive global quantitative council of Bank of America
October 07	Institute for Financial Management and Research (IFMR), Chennai. Taught a short course on Monte Carlo Methods in Finance to industry participants
Jan – April 07	Bank of America. Taught a course on Mathematical Finance to researchers and analysts at Bank of America Continuum Solutions at Mumbai
4/05 – 12/07	General Motors Research, India. In an R&D project, conducted research on modeling procurement auctions for project networks
12/05–6/07	Capital Metrics and Risk Solutions, Pune, India. Equity and financial analytics research firm. Helped develop cutting edge risk analysis and financial models
July 15, 2006	Indian School of Business, Hyderabad. Conducted a workshop on Computational Finance for industry participants
7/99 – 7/02	IBM India Research Lab, Delhi.

	Conducted research and development in the areas of e-commerce and communications networks
7/00 – 12/02	Delhi Transportation Corporation. Delhi bus traffic demand estimation and DTC bus route rationalization

Research grants

- Centre for Data, Learning and Data Science that I founded got fully funded by Safexpress Private Limited on August 2024
- Project with WHO, SEARO. Analysis of Potential Impact of Public Health and Social Measures 2021-22.
- Grant by ATE Chandra Foundation to enhance the IISc-TIFR Agent-Based Simulator. 2020-22
- Indo-Swedish Network Grant jointly with Dr. Pierre Nyquist and Prof. Henrik Hult at KTH, Sweden. 2020-22. Large deviations, rare-event simulation and machine learning: Importance sampling using neural networks
- MATRICS grant in Mathematical Sciences by Science and Engineering Research Board (SERB). 2020-22. Designing a Perfect Interview/Exam using Multi-Armed Bandit Methods

Patents awarded

- Distributed bid processing method for open-cry and descending price auctions. With Manish Gupta. Filed February 17, 2000. Awarded July 31, 2007. Number 7,251,630.
- System for optimal resource allocation and planning for hosting computing services. With Johara Shahabuddin; Kannan Balaji; Sanjiv Kapoor; Vishu Gupta; Ajay Chrungoo. Filed January 29, 2001. Awarded April 5, 2005. Number 6,877,035.
- A. Bassamboo, M. Gupta and S. Juneja. An Efficient Winner Determination Technique for Determining Winner Bids in Online Single Item, Multiple Units Auctions. Filed at USPTO.

Internships

6/92 - 9/92	IBM T.J. Watson Research Center, Yorktown Heights, NY.
6/91 - 9/91	Summer Intern. Performance Analysis Group, Computer Science Division
6/90 - 9/90	Bell Labs, Holmdel, NJ. Summer Intern. Operations Research Group

Outreach interactions

- Op-ed in Hindustan Times - Will Monsoons Become Impossible to Predict? With Tapio Schneider. Sept 25, 2026
- Article in The Week - Beyond Two Weeks: How AI Is Transforming Indian Monsoon Forecasting. With Manmeet Singh. Sept 14, 2026
- IIT, Stanford, COVID Modeling, and AI in Math.Decoded Podcast. Spotify
- Op-ed in Indian Express - Delhi’s winter pollution toolkit needs Al forecasts. With Manmeet Singh on July 22, 2026
- Subtext by Zerodha. Beyond Stock Picking: A Brief Introduction to Factor Investing in India. With Aman Foujdar, Atish Kumar & Nagpurnanand Prabhala · Aug 6, 2026
- Public Lecture at Chalmers AI Talk Series. Sweden. Shift, Scale and Restart Smaller Models to Estimate Larger Ones: Agent-based Simulators in Epidemiology. June 2, 2022.
- Popular Talk at Chai and Why event. Online. Covid-19 through the Mathematical Crystal Ball. April 4, 2021.

- City Scale Agent Based Simulators for the study of Covid-19 spread. CovidGyan WebGyan series. (link) June 18, 2020
- Invited by SynTalk (Mumbai, India) as a SynTalkr for #TIAU (The Infrequent And Unlikely, January 12, 2019).
  https://syntalk.wordpress.com/episodes/turn-five/tiau/ (website link)
- Popular Talk at Chai and Why event. Finding certainty in an uncertain world: A tale of tails. Oct. 2, 2016 at Prithvi Theatre.
- Invited to Speak at Mumbai Local, Junoon event on Modeling Uncertainty- A Tale of Tails January 15, 2016 at Kitabkhana, Mumbai
- Op-Ed piece supporting increased government investments in the IITs.
  http://www.indianexpress.com/news/pay-it-forward/514743/
- Popular talk at Chai and Why event Rocket Science and Ketchup Economics in Finance at Prithvi Theatre. Nov. 1, 2009.

Workshops co-organized

- exSPLOre events at Ashoka University January 2025, 26. Workshops on AI in Weather and Climate Modeling Sept 2024, 25, 26.
- Program on Data Science: Probabilistic and Optimization Methods. ICTS Bangalore. July 3-7, 2023, part 2 in 2025
- Workshop on Learning and Data Science. March 25-26, 2022. Ashoka University
- Workshop on Climate Studies. Hybrid. ICTS Bangalore. March 1-3, 2022
- Program on Advances in Applied Probability 2 (PAAPII), Online ICTS Bangalore, January 4-8, 2021
- Workshop on Learning Theory 2 at TIFR Jan 3-4, 2020
- Program on Advances in Applied Probability (PAAP) at ICTS Bangalore, August 5-17, 2019
- Workshop on Learning Theory at TIFR Jan 2-6, 2019
- Workshop Probability Day at TIFR in January 2018
- Three-day workshop on Applied Probability at TIFR in March 2017
- Two-day workshop on Learning and Related Probabilistic Applications at TIFR in February 2015
- Two-day workshop on Game Theory at TIFR in May 2013
- As part of ICTS, organized a two-week School in Mathematical Finance at TIFR in January 2012
- Member, Organizing Committee, International Colloquium on Perspectives in Fundamental Research, held at TIFR, March 2010
- Workshop on Stochastic Methods: Analysis and Algorithms. At TIFR, September 2009
- Summer School in Financial Mathematics (jointly with Prof. P. G. Babu) at IGIDR, April 2005

Key administrative roles

- Founding director. Safexpress Centre for Data, Learning and Decision Sciences. Ashoka University. February 2024 - present
- Dean, School of Technology and Computer Science. Jan 2017 to Sep 2021. September to December 2023
- Chair of selection committee for RM Tulpule Chair on Global Warming Research at TIFR

- Pan-TIFR Task force chair epidemiology studies, modelling and simulation activities. April 2020 to 2022.
- Chairperson, Faculty Induction and Recruitment Committee. School of Technology and Computer Science, TIFR. (2012 to 2016)
- Subject Board Convener, STCS, TIFR. 2005-06, 2012-13, Sept – Nov. 2016

Covid related reports

- Mittal, D., Juneja, S. and Agrawal, S., 2023. Shift, scale and restart smaller models to estimate larger ones: Agent based simulators in epidemiology. ACM SIGMETRICS Performance Evaluation Review, 50(4), pp.56-58.
- Juneja, S and Mittal, D., 2022, December. Modelling the Delta Covid-19 Wave in Mumbai. In *2022* Winter Simulation Conference (WSC) (pp. 581-592). IEEE

  .
- Mittal, D., Juneja, S. and Agrawal, S., 2022. Agent based simulators for Covid-19: Simulating larger models using smaller ones. *arXiv preprint arXiv:2209.02887*
- A. Eeshan, S. Juneja, D. Mittal, A. Noolkar, R. Saptharishi, P. Srivastava.
  Oxygen Planner for States in India STCS. https://www.tcs.tifr.res.in/
- Juneja, S., Mittal, D. 2021. Modelling the Second Covid-19 Wave in Mumbai https://arxiv.org/pdf/2105.02144.pdf
- Malani, A., Shah, D., Kang, G., Lobo, G.N., Shastri, J., Mohanan, M., Jain, R., Agrawal, S., Juneja, S., Imad, S. and Kolthur-Seetharam, U., 2021. Seroprevalence of SARS-CoV-2 in slums versus non-slums in Mumbai, India. **The Lancet Global Health**, *9*(2), pp.e110-e111. (link) Medrxiv version https://www.medrxiv.org/content/10.1101/2020.08.27.20182741v1
- Harsha, P., Juneja, S., Mittal, D. and Saptharishi, R. 2020. COVID-19 Epidemic in Mumbai: Long term projections, full economic opening, and containment zones versus contact tracing and testing. Preprint. (updated_October pdf) https://arxiv.org/abs/2011.02032 (Early September pdf)
- S. Agrawal, S. Bhandari, A. Bhattacharjee, A. Deo, N. Dixit, P. Harsha, S. Juneja, P. Kesarwani, A. Swamy, P. Patil, N. Rathod, R. Saptharishi, S. Shriram, P. Srivastava, R. Sundaresan, N. K. Vaidhiyan, and S. Yasodharan (2020). City-Scale Agent-Based Simulators for the Study of Non-Pharmaceutical Interventions in the Context of the COVID-19 Epidemic. **Journal of the Indian Institute of Science**, 1-39.
- Mumbai-Serosurvey Technical report-NITI_BMC-Round-2
- Mumbai-Serosurvey Technical report-NITI
- P Harsha, S Juneja, P Patil, N Rathod, R Saptharishi, A. Y. Sarath, S Sriram, P Srivastava, R Sundaresan, N K. Vaidhiyan. 2020. COVID-19 Epidemic Study II: Phased Emergence From the Lockdown in Mumbai. http://arxiv.org/abs/2006.03375

Publications

2026

- Avireddy, A., Singh, M., Singh, S., Raj, A., Dhanuka, S., Mukhopadhyay, P. and Juneja, S.K. 2026. Steering Diffusion Priors with Sparse Observations for High-Resolution Temperature Downscaling. **NeurIPS 2026** Workshop: Tackling Climate Change with Machine Learning. (pdf)
- Raj, A., Dhanuka, S., Singh, M., Mukhopadhyay, P., Goswami, B. and Juneja, S.K. 2026. IndiaS2S-Bench: A Comprehensive Benchmark of Subseasonal to Seasonal Forecasting over India. **NeurIPS 2026** Workshop: Tackling Climate Change with Machine Learning. (pdf)
- Dhanuka, S., Iyer, S.R., Singh, M., Mohapatra, M., Mukhopadhyay, P. and Juneja, S.K. 2026. Learning Ensemble Mixtures from Observations and Adapting Online with Theoretical Bounds. **NeurIPS 2026** Workshop: Tackling Climate Change with Machine Learning. (pdf)
- Raj, A., Singh, M., Dhanuka, S., Goswami, B., Mukhopadhyay, P., Juneja, S.K. and Mohapatra, M. 2026. Calibrating AI Subseasonal to Seasonal (S2S) Forecasts with Lightweight Regional Adapters. **NeurIPS 2026** Workshop: Tackling Climate Change with Machine Learning. (pdf)
- Kumar, O., Dey, S., Iyer, S.R. and Juneja, S. 2026. Directionally Correct Views in Portfolio Optimization. **NeurIPS 2026** Workshop: Second Workshop on ML×OR. (pdf)
- Gupta, R. and Juneja, S. 2026. A Splitting Method for SDE Terminal-Law Estimation. **NeurIPS 2026** Workshop: Second Workshop on ML×OR. (arxiv)
- Dhanuka, S., Iyer, S., Singh, M., More, M., Gupta, R., Gupta, D., Mukhopadhyay, P. and Juneja, S., 2026. AdaWeather: Adaptively Mixing Probabilistic Weather Forecasts with Logarithmic Regret. arXiv preprint arXiv:2606.02663.
- Deep, V., Bassamboo, A. and Juneja, S., 2026. Asymptotic optimality theory of confidence intervals of the mean. Proceedings of **AISTATS 2026**, PMLR 300. *arXiv:2501.19126*.
- Iyer, S.R., Gupta, D., Gupta, R., Mandal, H., Bandyopadhyay, A., Bassamboo, A., Gupta, V., Juneja, S. 2026. Fundamental limits for weighted empirical approximations of exponentially tilted distributions. Proceedings of **AISTATS 2026**, PMLR 300.
- Luitel, S., Singh, M., Durkee, J., Fahad, A.A., Sudharsan, N., Singh, P., He, C., Kamath, H., Yang, Z.L., Halder, K. and Juneja, S., 2026. AirCast-SR: A Foundation Model for Kilometer-Scale Atmospheric Super-Resolution via Latent Consistency Diffusion. arXiv preprint arXiv:2605.26130.
- Luitel, S., Singh, M., Mukhopadhyay, P., Juneja, S. and Dhanuka, S., 2026. Evaluating the Fidelity of GraphCast AI Forecasts for the Indian Summer Monsoon: A Climatological Assessment Against ERA-5 Reanalysis and IMERG Observations. arXiv preprint arXiv:2607.11905.
- Bandyopadhyay, A. and Juneja, S., 2026. Game of arrivals to a two queue network with heterogeneous customer routes. **ACM SIGMETRICS Performance Evaluation Review**, 53(4), pp.97-102.
- Mandal, H., Gupta, D., Gupta, R., Iyer, S.R., Bandyopadhyay, A., Bassamboo, A., Gupta, V., Juneja, S. 2026. Generating DDPM-based Samples from Exponentially Tilted Distributions. Submitted. arXiv:2604.03015.

2025

- Pagare, T., Bandyopadhyay, A. and Juneja, S. 2025. Optimal Algorithms for Bandit Learning in Matching Markets. Dynamics at the Frontiers of Optimization, Sampling, and Games – **NeurIPS 2025** Workshop (arxiv)
- Jain, A., Batra, J., Narula, A., Rajeevan MN, and Juneja, S. 2025. Deep learning for short-range monsoon rainfall forecast using ground truth rainfall data. Tackling Climate Change with Machine Learning **NeurIPS 2025** Workshop. (paper)
- Ahn, D., Juneja S., Pagare, T. and Shreyas, S. Data-driven estimation of tail probabilities under varying distributional assumptions. **Winter Simulation Conference** (WSC). IEEE, 2025.
- Dhanuka, S., Singh, M. and Juneja, S. 2025. Finetuning a Neural Weather Model to India Rainfall Accepted to ACM IKDD CODS 2025 13th International Conference on Data Science AI for Sciences Track.
- Foujdar, A., Juneja, S., Kumar, A. and Prabhala, N. 2025. Risk Factors for the Indian Equity Market: Statistics, Visualization, and an Interactive Tool. Preprint.

2024

- Bandyopadhyay, A., Juneja, S. and Agrawal, S., 2024. Optimal top-two method for best arm identification and fluid analysis. ***Advances in Neural Information Processing Systems***, *37*, pp.66568-66646.
- Deep, V., Bassamboo, A. and Juneja, S.K., 2024, July. Asymptotically Optimal and Computationally Efficient Average Treatment Effect Estimation in A/B testing. In ***International Conference on Machine Learning*** (pp. 10317-10367). PMLR.
- Hult, H., Jain, A., Juneja, S., Nyquist, P. and Vijayan, S., 2024, December. A Deep Learning Approach for Rare Event Simulation in Diffusion Processes. In *2024* ***Winter Simulation Conference*** *(WSC)* (pp. 2559-2570). IEEE.
- Bhattacharjee, A. and Juneja, S., 2024, December. Selecting the Safest Design in Rare Event Settings. **In *2024 Winter Simulation Conference*** *(WSC)* (pp. 3403-3414). IEEE.
- Narula, A., Jain, A., Batra, J. and Juneja, S., 2024. Comparing skill of historical rainfall data-based monsoon rainfall prediction in India with NCEP-NWP forecasts. *arXiv preprint arXiv:2402.07851*.

2023

- Bhattacharjee, A., Vijayan, S. and Juneja, S., 2023, July. Best arm identification in rare events. In **Uncertainty in Artificial Intelligence** (pp. 163-172). PMLR.
- Mittal, D., Juneja., S. and Agrawal, S. 2023. Agent based simulators for epidemic modelling: Simulating larger models using smaller ones. Proc. 4th International **AAMAS Workshop on Autonomous Agents for Social Good (AASG 2023)**.
- Agrawal, S., Juneja, S., Shanmugam, K. and Suggala, A.S., 2023. Optimal Best-Arm Identification in Bandits with Access to Offline Data. *arXiv preprint arXiv:2306.09048*
- Mittal, D., Juneja, S. and Agrawal, S., 2023. Shift, scale and restart smaller models to estimate larger ones: Agent based simulators in epidemiology. **ACM SIGMETRICS Performance Evaluation Review**, 50(4), pp.56-58.
- Bassamboo, A., Deep, V., Juneja, S. and Zeevi, A., 2023. Learning to Ask the Right Questions: A Multi-armed Bandits Approach. *Available at SSRN 4375185*.
- Bandyopadhyay, A. and Juneja, S. 2023. Game of arrivals at a two-queue network with heterogeneous customer routes. Preprint.

2022

- Deep, V., Bassamboo, A., Juneja, S. and Zeevi, A., 2022, Exact Optimal Fixed Width Confidence Interval Estimation for the Mean. In *2022* **Winter Simulation Conference** (WSC) (pp. 713-723). IEEE.
- Juneja, S and Mittal, D., 2022, Modelling the Delta Covid-19 Wave in Mumbai. In *2022* **Winter Simulation Conference** (WSC) (pp. 581-592). IEEE
- Juneja, S., 2022. Learning the queue arrivals game equilibrium. **Queueing Systems**, 100 (3-4), pp.533-535.
- Mittal, D., Juneja, S. and Agrawal, S., 2022. Agent based simulators for Covid-19: Simulating larger models using smaller ones. *arXiv preprint arXiv:2209.02887*

2021

- Moka, S. B., S. Juneja and M. R. H. Mandjes. 2021. Rejection and Importance Sampling based Perfect Simulation for Gibbs Hard Sphere Models. arXiv:1705.00142. **Advances in Applied Probability**, *53*(3), pp.839-885.
- Agrawal, S., Juneja, S. and Koolen, W.M., 2021. Regret Minimization in Heavy-Tailed Bandits. **Conference on Learning Theory** (COLT) 2021. 134. Pp. 1-37. https://arxiv.org/pdf/2102.03734.pdf.
- Agrawal, S., Koolen, W.M. and Juneja, S., 2021. Optimal Best-Arm Identification Methods for Tail-Risk Measures. **Advances in Neural Information Processing Systems**, 34. https://arxiv.org/abs/2008.07606
- Malani, A., Shah, D., Kang, G., Lobo, G.N., Shastri, J., Mohanan, M., Jain, R., Agrawal, S., Juneja, S., Imad, S. and Kolthur-Seetharam, U., 2021. Seroprevalence of SARS-CoV-2 in slums versus non-slums in Mumbai, India. **The Lancet Global Health**, 9(2), pp.e110-e111. (link) Medrxiv version https://www.medrxiv.org/content/10.1101/2020.08.27.20182741v1
- Anand Deo, Sandeep Juneja. 2021. Credit Risk: Simple Closed-Form Approximate Maximum Likelihood Estimator. **Operations Research** 69(2):361-379. https://doi.org/10.1287/opre.2020.2029
- Agrawal, S., Juneja, S. and Koolen, W.M., 2021. Regret minimization in risk averse bandits. In 2021 Seventh **Indian Control Conference** (ICC), pp. 195-200. IEEE, 2021.

2020

- S Agrawal, S Juneja and P Glynn. 2020. Optimal delta-correct best arm selection for heavy-tailed distributions. **Algorithmic Learning Theory** (ALT 2020). Pp 61-110. https://arxiv.org/abs/1908.09094.
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- S. Juneja, P. Shahabuddin. 1992. `Fast Simulation of Markovian Reliability/Availability Models with General Repair Policies', Proc. of the twenty second Int'l Symp. on Fault-Tolerant Computing, IEEE Press, 150-159.

Selected Invited Talks (2003 onwards)

- AI for Math Workshop, TIFR Mumbai August 6, 2026. Incorporating views in Portfolio optimization and generating data from tilted distributions
- Indian Institutional Quant Conference. Organized by LAQSA. Gurgaon July 17, 2026. India Factor Library & Portfolio Optimization
- 2026 Summer Simulation Workshop, 12-13 June 2026 at Tianjin University, China. Asymptotic optimality theory of confidence intervals of the mean
- Conference on Mathematical Finance, IISER Pune, June 11-13, 2026. Flipping Coins to Win!
- Shanghai Jin Tao University. June 18, 2026. Sequential Learning in Bandits
- Nanjing University, June 17 2026. Flipping Coins to Win!
- Plaksha University. Nov 27, 2025. Sequential Learning in Bandits
- IIT Bombay. Nov 2, 2025. Generating samples from exponentially tilted distributions via diffusions.
- CMI-BIRS Workshop on Mathematical and Computational Foundations of Climate Modelling. Aug 16-20, 2025. Comparing skill of historical rainfall data based monsoon rainfall predictions with NWP forecasts
- AI Summer School, IIIT Delhi, August 2025. Fluid approach to a few pure explorations multi-armed bandit problems
- CMI-BIRS 3rd Workshop on Game-theoretic Statistics and Sequential, Anytime-Valid Inference (SAVI) June 29 to July 4, 2025 at the Chennai Mathematical Institute - Best arm identification in presence of offline data - top two methods, fluid analysis
- At ISB, March 13, 25. Spoke on - Flipping coins to win!
- At Capacity Building Program at Industrial Development and Policies for Realising Viksit Bharat@2047, on Sept. 12, 2024 - Sequential Learning Methods
- Selecting the best arm – optimal algorithm based on fluid analysis, in presence of offline data. Keynote address. INFORMS Simulation Society Research Workshop, Hong University of Science and Technology. June 24-26, 2024.
- Member, Panel discussion on AI and Simulation Research. I-Sim Research Workshop, Hong University of Science and Technology. June 24-26, 2024.
- Shift, Scale and Restart Smaller Country Level Models to Estimate Larger Ones: Agent-based Simulators for Covid Modelling. CUHK, June 27, 2024.
- Selecting the best arm – optimal algorithm based on fluid analysis. Kellogg Operations Seminar, Northwestern University. June 13th 2024
- Flipping Coins to Win. International Conference on Stochastic Calculus and Applications to Finance. IIT Madras. June 3-5, 2024.
- Flipping Coins to Win. Keynote address. Workshop on Economics and Computation. IIIT Delhi. April 26-28.
- Modelling and data challenges, Challenges in Epidemiological Modelling in India. IASc Discussion Meeting. Coorg, Feb 18-22, 2024.
- Flipping Coins to Win. Wednesday Colloquium. TIFR Mumbai Feb 14, 2024.

- Best arm identification in multi-armed bandits – optimal algorithm based on fluid analysis, ICTS Workshop on Networks. Feb 1, 2024.
- Best arm identification in multi-armed bandits – optimal algorithm based on fluid analysis, Advances in Probability Theory and Applications to Systems Modeling, IIT Bombay Feb 2-3, 2024
- Sequential learning – a fluid analysis. ICAAP and SP 2024, Thissur. January 18-20.
- Sequential learning in a stochastic multi-armed bandit framework, IIT Kanpur Economic Sciences Department. Oct. 5, 2023.
- Trends in Operations Research. Chief Guest IEOR Department Convocation August 19, 2023, IIT Bombay.
- Guided top-2 approach to optimally solve the best arm identification problem. INFORMS Applied Probability Conference 2023. Nancy, France June 28-30. 2023.
- Agent based simulators for epidemic modelling: Simulating larger models using smaller ones. INFORMS Applied Probability Conference 2023. Nancy, France June 28-30. 2023.
- Sequential learning in a stochastic multi armed bandit framework. Workshop on Artificial Intelligence and Machine Learning. Ashoka University. May 1-3, 2023.
- Sequential learning in a stochastic multi armed bandit framework. Bangalore Theory Seminar Series, CSA, IISc. April 5, 2023
- Sequential learning in a stochastic multi armed bandit framework. Research Week with Google 2023. January 29-31. 2023.
- Sequential learning in a stochastic multi armed bandit framework. International Symposium on Applied Optimization and Game Theoretic Models for Decision Making. ISI Delhi. Feb 1-3, 2023
- Conducted a 5 hour short course on Sequential learning under uncertainty through multi armed bandit framework. At Workshop in Probability and Statistics, Dept. Mathematics, IIT Bombay. Jan 3-6, 2023.
- Optimal Best-Arm Identification Methods for Tail-Risk Measures. CODS COMAD Jan 4-7, 2023, IIT Bombay.
- Agent-based Simulators for Covid Modelling. Symposium on Epidemic Modelling. December 14, 2022. IIT Madras
- Agent-based City Simulators and Covid Modelling of Mumbai with efficient algorithmic enhancements. Keynote address at International Conference on Data, Decision and Systems (ICDDS 2022) December 2-3, 2022
- Learning to ask the right questions using multi-armed bandits. Invited talk at Ashoka University. Workshop on Computation and Economics. Sep 9-11, 2022
- Shift, Scale and Restart Smaller Models to Estimate Larger Ones: Agent-based Simulators for Covid Modelling. Online, GPCE Seminar Series, Biocomplexity Institute, University of Virginia. July 27, 2022.
- Stochastic Multi Armed Bandits and Heavy Tails. Chalmers University, CS. June 3, 2022.
- Stochastic Multi Armed Bandits and Heavy Tails. KTH Royal Institute of Technology. May 30, 2022.
- Stochastic Multi Armed Bandits and Heavy Tails. Stochastic Networks, Applied Probability, and Performance (SNAPP) Seminar Series. Oct 18, 2021

- Shift, Scale and Restart Smaller Models to Estimate Larger Ones: Agent Based Simulators in Epidemiology. University of Mannheim (online). Plenary talk at Monte Carlo Methods and Applications (MCM) 2021. August 16-20, 2021.
- Modelling Covid Spread in a City through Simulations and some Algorithmic Enhancements. Workshop on COVID-19: Data and Modelling. University of Hyderabad. July 9-10, 2021
- Shift, scale and reset smaller models to estimate larger ones: Agent based simulators in epidemiology. 2021 INFORMS Simulation Society Workshop (I-Sim). Online, Penn State University. June 20-23, 2021.
- Shift and Scale Approach to Speed up Epidemiological Agent Based Simulation Models. International Symposium on Computational Operations Research and Algorithmic Game Theory March 29-31, 2021
- Partition identification using multi-armed bandits for heavy-tailed distributions. IEEE SPS Seminar Series on Optimization and Learning. IIT Kanpur, Nov 27, 2020.
- Mumbai Sero Survey: Facts and interpretations. TNQ Janelia India Covid-19 Seminar (jointly with Prof. Kolthur), October 16
- Mumbai Sero Survey: Facts and interpretations. Special TIFR Colloquium, August 7, 2020 (jointly with Prof. Kolthur)
- Partition Identification using Multi-Armed Bandits for Heavy-Tailed Distributions. At Recent Trends in Convex Optimization; Theory, Algorithms and Applications (RTCOTAA-2020). 29th to 31st Oct. 2020 at Department of Mathematics, IIT Patna
- Cross-sectional SARS-CoV2 Sero-surveillance study in Mumbai: Facts and Interpretations. SOLIDARITY II, WHO
- Mumbai Sero Survey presentation to COVID Epi-Modelling Team, Gates Foundation. August 13 (jointly with Prof. Kolthur)
- Partition identification using multi-armed bandits for heavy-tailed distributions. Seminar on Optimization, Variational Analysis and Applications. Speaker and Chief Guest. Institute of Science, BHU. Feb. 2-4, 2020.
- Multi-Armed Bandits. International Conference on Operations Research and Decision Sciences (ICORDS 2019). IIM Vishakhapatnam. Keynote speaker. Dec. 28-30, 2019
- Partition identification using multi-armed bandits for general distributions. Workshop on Statistical Finance. CMI. Dec 19-21, 2019
- Learning Optimal Sequential Decisions using Stochastic Multi-Arm Bandit Methods. Keynote Lecture. CORE 2019. IGIDR PhD. Colloquium, November 5-8, 2019
- Discriminative Learning via Adaptive Questioning. INFORMS Applied Probability Conference. Queensland, Australia. July 3-5, 2019
- Discriminative Learning via Adaptive Questioning. Workshop on Stochastic Systems and Network Algorithms. IIT Bombay. March 22-24, 2019.
- Sample complexity of partition identification using multi-armed bandits with applications to nested Monte Carlo. Keynote speaker in workshop on Advances and challenges in Monte Carlo Methods at The University of Queensland. Nov. 30, 2018.
- Sample complexity of partition identification using multi-armed bandits. Keynote speaker in 51st Annual convention of ORSI and International Conference at IIT Bombay. Dec. 17, 2018.
- Multi Armed Bandits, a short course. Winter School on Stochastic Methods for Uncertainty Quantification and Sensitivity Analysis of Complex Models organized by IFCAM at IISER Kolkata. December 4, 2018.

- Invited talks at INFORMS Annual Meet 2018 on - Optimal Hardness of Questions in Static and Interactive Exams and Partition Identification using Multi-armed Bandit Methods with Applications to Financial Risk
- Partition identification using multi-armed bandit methods with applications to financial portfolio risk. Colloquium at ISYE Stochastics group, Georgia Tech., May 14, 2018.
- Partition identification using pure exploration multi-armed bandits. EE Department, IITM, August 6, 2018
- Partition identification using multi-armed bandit methods with applications to financial portfolio risk. Workshop on Mathematical Finance, July 12-16,2018 IISER Pune
- Queueing Games: To Wait or to be Late. Colloquium at CEBS, Mumbai. January 18, 2018
- Perfect Sampling for Gibbs Processes with a focus on Hard-sphere Models. At International Symposium on Operations Research and Game Theory: Modeling and Computation. January 9-11, 2018. ISI Delhi
- Financial Credit Risk. Seminar organized by Mechanical Engineering Society (MES) at IITD. Jan 10, 2018
- Dynamic Portfolio Credit Risk: Calibration, Modelling and Analysis. Ninteenth INFORMS Applied Probability Society Conference, July 10-12. At Northwestern University
- Rest in the lounge or directly join the queue. Ninteenth INFORMS Applied Probability Society Conference, July 10-12. At Northwestern University
- Selecting the best population using large deviations and multi-armed bandit methods. At IITB, Seminar at EE Deptt. October 9, 2017
- Portfolio Credit Risk: Simple Closed Form Approximate Maximum Likelihood Estimator and Related Issues. Colloquium at ICTS, Bangalore. September 4, 2017
- Credit Risk: Simple Closed Form Approximate Maximum Likelihood Estimator, comparison and analysis. IGIDR-NESL workshop on Mathematics and Economics, Sept 2, 2017
- Selection of the Best System using large deviations, and multi-arm Bandits. At a program on Large deviation theory in statistical physics: Recent advances and future challenges. At ICTS. August 23, 2017
- Acceptance-rejection and Importance Sampling Methodologies for Perfect Sampling from Gibbs Point Processes. At Ramanujan Math Society Conference. June 25, 2017 Belagavi
- Credit Risk Measurement. At Applications of Mathematics Workshop at IITGN on May 16, 2017
- Rest in the lounge or directly join the queue. 2017 Symposium on Mathematical Programming and Game Theory. ISI Delhi. January 9-11, 2017
- A new approach to calibrating credit risk. Reserve Bank of India. SRU-CAFRAL Seminar. Feb. 22, 2017
- Dynamic Portfolio Risk Measurement. At Conference on Statistical Methods in Finance, at CMI, Dec. 2016.
- Dynamic Portfolio Risk Measurement. At Finance and Stochastics Day 2016, Imperial College, UK, 13 October 2016.
- Modeling Credit Risk. Valedictory address at Workshop on Big data, Economics and Finance, at Presidency College on 26 October 2016
- Rest in Lounge or Wait in Queue. Workshop on Congestion Games. Held at Institute for Mathematical Sciences, National University of Singapore. December 16, 2015
- Strategic arrivals to queues. Keynote speaker at the XIX Annual Conference of Society of

Operations Management. Held at IIM Calcutta. December 12, 2015

- Ordinal optimization - Empirical large deviations rate estimators, and multi-armed bandit methods. At Seminaire Bachelier Paris. Held at Henri Poincare Institute. May 29, 2015
- Large deviations, selecting the best population and multi-armed bandit methods. At International Conference on Applied Probability and Computational Methods in Applied Sciences. Held at Shanghai Centre for Mathematics, Fudan University. November 2, 2015
- Ordinal optimization - Empirical large deviations rate estimators, and multi-armed bandit methods. At the workshop Applied Probability Frontiers: Computational and Modeling Challenges. At Banff International Research Station, Canada. June 1, 2015
- Multi Armed Bandit Sampling in Nested Simulation for Financial Portfolio Risk Measurement. Keynote speaker at 4th IIMA International Conference on Advanced Data Analysis, Business Analytics and Intelligence. Held at IIM Ahmedabad. April 11, 2015
- Ordinal optimization and multi-armed bandits, Portfolio risk measurement using multiarmed bandit methods, At NMI conference on Non-convex Optimization for Machine Learning. IIT Bombay. June 15 and 16, 2015
- Large deviations, selecting the best population and multi-armed bandit methods, Department of Mathematics Colloquium, IIT Bombay, October 4, 2015
- Efficient Rare Event Simulation Algorithms for Heavy Tailed Processes, Symposium on Learning, Algorithms and Complexity, January 5-9, 2015, Indian Institute of Science, Bangalore, India
- Optimization in Simulation and Pure Exploration Multi-Armed Bandit Methods. At Workshop on Applied Optimization Models and Computation. January 28-30, 2015. ISI Delhi
- Multi-armed Bandit sampling in Nested Portfolio Risk Measurement. NUS-University of Tokyo Workshop on Quantitative Finance. Sept. 25-26, 2014
- Ordinal Optimization and Multi-armed Bandit Techniques. INFORMS Annual Meet, November 2014 in San Francisco
- Ordinal optimization in simulation and pure exploration multi-armed bandit methods. Conference on Stochastic Analysis and Applications, Dept. of Mathematics, IISc, September, 8 - 11, 2014
- Rare event simulation of heavy tailed random walks - A new approach. Stochastic Networks Conference at CWI Amsterdam, June 23-27, 2014
- Rare event simulation of heavy tailed random walks - A new approach. IIT Guwahati Mathematics Seminar Series. March 25, 2014
- Concert Queue Arrivals Game: An Overview. Seminar at Department of Applied Mathematics, Ecole Polytechnique, Palaiseau Cedex, France, November 14, 2013
- On Concert Queue Arrivals Game: An Overview. Seminar at Mathematics Department, IISc Bangalore, December 18, 2013
- State Independent Important Sampling for Regularly Varying Distributions. INFORMS Annual Meet. October 2013
- Asymptotic Comparison of Popular Approaches to Price Bermudan Options. INFORMS Annual Meet. October 2013
- Rare Event Simulation of Heavy Tailed Random Walks. Colloquium at University of Southern California EE Dept. Oct. 2013
- The Concert Queueing Problem: Processor Sharing Regime. ISI Delhi, International Symposium on Applied Optimization and Game-Theoretic Models January 9-11, 2013

- The Concert/Cafeteria Queueing Game. ICERM, Brown University, November 21, 2012.
- The Concert/Cafeteria Queueing Game. Operations Management Seminar at Sloan School, MIT, November 5, 2012
- Monte Carlo Methods in Finance. ASET Colloquium, TIFR, September 21, 2012
- Computational finance. TCS Innovations Research Lab, Hyderabad, March 20, 2012: Conducted a short course
- Monte Carlo Simulation and Modelling. National Workshop and Training Programme on Differential Equations and Mathematical Modelling at Lady Shri Ram College, Delhi. Feb. 9, 2012. Plenary speaker
- Efficient Estimation of Density and Probability of Large Deviations of Sum of IID Random Variables INFORMS Winter Simulation Conference, Phoenix, Arizona, December 2011
- Ordinal optimization: A nonparametric framework INFORMS Winter Simulation Conference, Phoenix, Arizona, December 2011
- Monte Carlo Methods in Finance, TACTICS Symposium 2011 (Hosted by Tata Consulting Services Research) December 2011
- The Concert Queuing Game with a Finite Homogeneous Population. University of Amsterdam, November 3, 2011
- The Concert/Cafeteria Queuing Game with a Fluid/Finite Population. University of Twente, November 7, 2011
- Multidimensional Fourier Inversion using Importance Sampling in Rare Event Simulation and Finance. The Applied Probability Society Conference, July 2012, Stockholm
- The concert queuing game: Stochastic system with homogeneous users. The Applied Probability Society Conference, July 2012, Stockholm
- The concert queueing game: To wait or to be late University of Bristol, May 20, 2011
- Newton Institute, Cambridge, UK. June 16, 2010. Presented: The concert queueing game: to wait or to be late
- Maxwell Institute for Mathematical Sciences, Heriot Watt University, Edinburgh, September 29, 2010. Presented: Estimating Mean of Non-Linear Function of a Conditional Expectation with Applications to Portfolio Risk Measurement
- Maxwell Institute for Mathematical Sciences, Heriot Watt University, Edinburgh, September 29, 2010. Presented: Concert Queuing Game: To Wait or To be Late
- INFORMS Annual Meeting 2010. Texas Austin. Presented: Entropy Approach for Incorporating Fat-tailed Constraints in Financial Models
- INFORMS Annual Meeting 2010. Texas Austin. Presented: The Concert Queuing Arrivals Game: Finite Customer Analysis
- Stanford University, Operations Research Seminar, November 15, 2010: Presented: The Concert Queuing Game
- IIT Bombay, IEOR Dept., November 4, 2010. Presented: The Concert Queuing Game: To Wait or To be Late
- INFORMS Winter Simulation Conference, Baltimore, Maryland, December 2010. Conducted a tutorial on: Monte Carlo Methods in Finance: An Introductory Tutorial
- INFORMS Winter Simulation Conference, Baltimore, Maryland, December 2010. Presented: Multidimensional Fourier Inversion using Importance Sampling with Application to Option Pricing

- December 19th-22nd, 2010 at Quantitative Finance Workshop at ITM Institute of Financial Markets, Mumbai, INDIA. Presented: Introduction to Financial Mathematics
- Workshop on Financial Mathematics at Dept. of Mathematics, Indian Institute of Science, Bangalore, May 10 -15, 2009. Presented: Computational issues in pricing multi-dimension American and European Options
- Mumbai-Pune Soft Matter meeting, TIFR. June 6, 2009. Presented: Rare Event Analysis and Simulation.
- Applied Probability Conference at Cornell University, July, 2009. Presented:
  - The Concert Queueing Problem: A Game of Arrivals
  - Incorporating views in Mathematical Models using Entropy Approach
  - Simulation Estimation of the Mean of a Non-linear Function of a Conditional Expectation
- Conference on Probability and Stochastic Process at ISI Delhi, Nov. 26-28. Presented: The Concert Queueing Problem: A Game of Arrivals
- CAFRAL-IFMR-IISc-CMI conference on Application of Advanced Computational and Mathematical Methods in Finance held on September 26-27 at Parvathy Hall, IFMR Campus, Chennai. Presented: Advances in Risk Measurement Techniques
- International Conference on Methods and Models in Computer Science held in Jawaharlal Nehru University on Dec. 14, 2009. Presented: Portfolio Risk Measurement
- Winter Simulation Conference at Miami, December, 2008. Presented:
  - Optimizing Portfolio Tail Measures: Asymptotics and Efficient Simulation Optimization
  - A Large Deviations View of Asymptotic Efficiency for Simulation Estimators
- Conference on 'Efficient Monte Carlo: From Variance Reduction to Combinatorial Optimization' held at Sandbjerg Estate, Sønderborg, Denmark 14-18 July 2008. Presented: Nested Simulation in Risk Management
- Colloquium, IIM Bangalore on November 14, 2008. Presented: Extremal Dependence in Portfolio Credit Risk Modeling
- International Conference on Modeling, Computation and Optimization, Indian Statistical Institute, New Delhi, January 09-10, 2008. Presented: Nested Simulation in Portfolio Risk Measurement.
- International Workshop on Rare Event Simulation, The University of Nice, Azura Coast, France, April 30-May 04, 2007. Presented: Simultaneous Estimation of Many Rare Events
- Workshop on Rare Events in Communication Networks, Heriot Watt University, Edinburgh, UK July 02-07. 2007. Presented: Estimating Rare Events Involving Heavy Tailed Random Variables with Asymptotically Zero Relative Error
- INFORMS Applied Probability Conference, EURANDOM, The Netherlands, July 09-11, 2007. Presented:
  - Uniformly Efficient Importance Sampling for the Tail Distribution of Sums of Random Variables
  - Minimizing Tail Probabilities: Asymptotic Analysis and Efficient Simulation optimization
- International Conference on Stochastic Processes and Applications, Indian Institute of Science, Bangalore, July 16-21, 2007. Presented: Simultaneous Estimation of Many Rare Events
- Conducted a short course on Computational Finance for the industry participants at the Indian School of Business, July 2006

- Winter Simulation Conference at Monterey, California, December 3-6, 2006. Presented: Efficient Simulation Techniques for Portfolio of CDOs.
- Gave a tutorial at the Tenth Annual Society for Operations Management Conference held at IIM Ahmadabad, December 21-23, 2006. Presented: Introduction to Monte Carlo methods
- Delivered the First Perwez Shahabuddin Memorial Lecture held at IIT Delhi, February 27, 2006. Presented: Rare Event Simulation
- Gave series of lectures on Monte Carlo Methods in a Summer School on Mathematical Finance held at IGIDR, Mumbai. April 2005
- Winter Simulation Conference at Orlando, Florida, December 7-9, 2005. Presented: Function-Approximation-based Control Variates for Pricing American Options
- Summer Research Conference at the Indian School of Business, Hyderabad, August 7-9, 2005. Presented: Portfolio Credit Risk under Extremal Dependence
- Symposium on Current Topics in Operations Research, November 12, 2005. Organized by IEOR Department in IIT Bombay. Presented: Monte Carlo Methods in Finance
- Winter Simulation Conference at Washington D. C. December 4-6, 2004. Presented:
  - Function-Approximation-based Importance Sampling for Pricing American Options
  - A Large Deviations Perspective on Ordinal Optimization
- Workshop on Monte Carlo Methods, Coorg, Nov. 28-Dec. 3, 2004. Presented a series of lectures on Importance Sampling
- Society of Mathematical Sciences Annual Meeting 2004. Held in October 11, 2004 in Delhi University. Presented: Monte Carlo Methods in Finance
- INFORMS (Institute for Operations Research and Management Sciences Annual meet 2003) at Atlanta October 19-22, 2003. Presented:
  - Adaptive Importance Sampling for Markov Chains using Stochastic Approximation
  - Optimizing QoS of Composite Web Services modeled as PERT Networks
- Sixth International Conference of the Association of Asia Pacific Operations Research Society, December 2003 in Delhi. Presented:
  - Selecting the best design amongst many alternatives via simulation using large deviations theory
  - Adaptive Importance Sampling for Markov Chains
- International Conference on Operations Research with Economic and Industrial Applications 2004. Presented: Computationally Efficient Winner Determination Techniques for Internet Multi-Unit Auctions.
- Workshop on Applied Probability. Held on April 18, 2003 at TIFR. Presented: A large deviations perspective on ordinal optimization.